

# BAAZ & UDAY JKBOSE

2026 CLASS 11<sup>TH</sup>

## BIOMOLECULES

LECTURE-2

ZOOLOGY

DAVOOD SIR (CEO of NG)

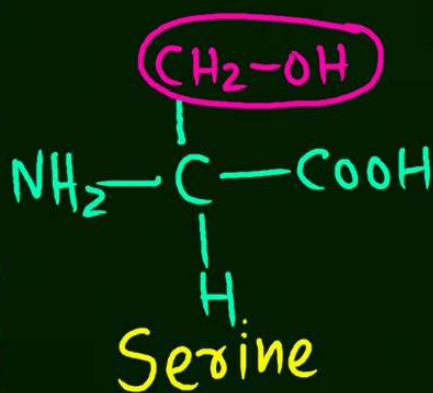
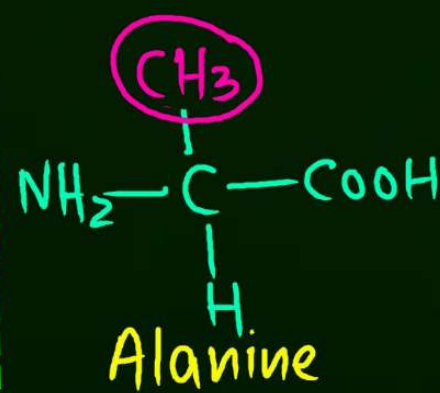
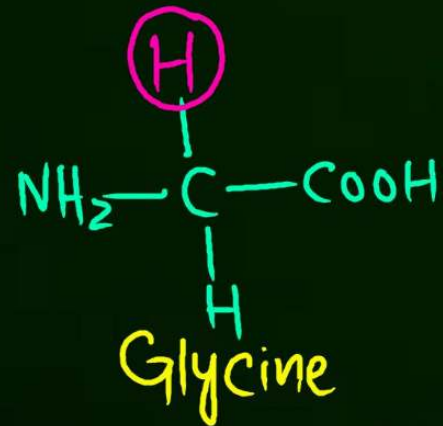
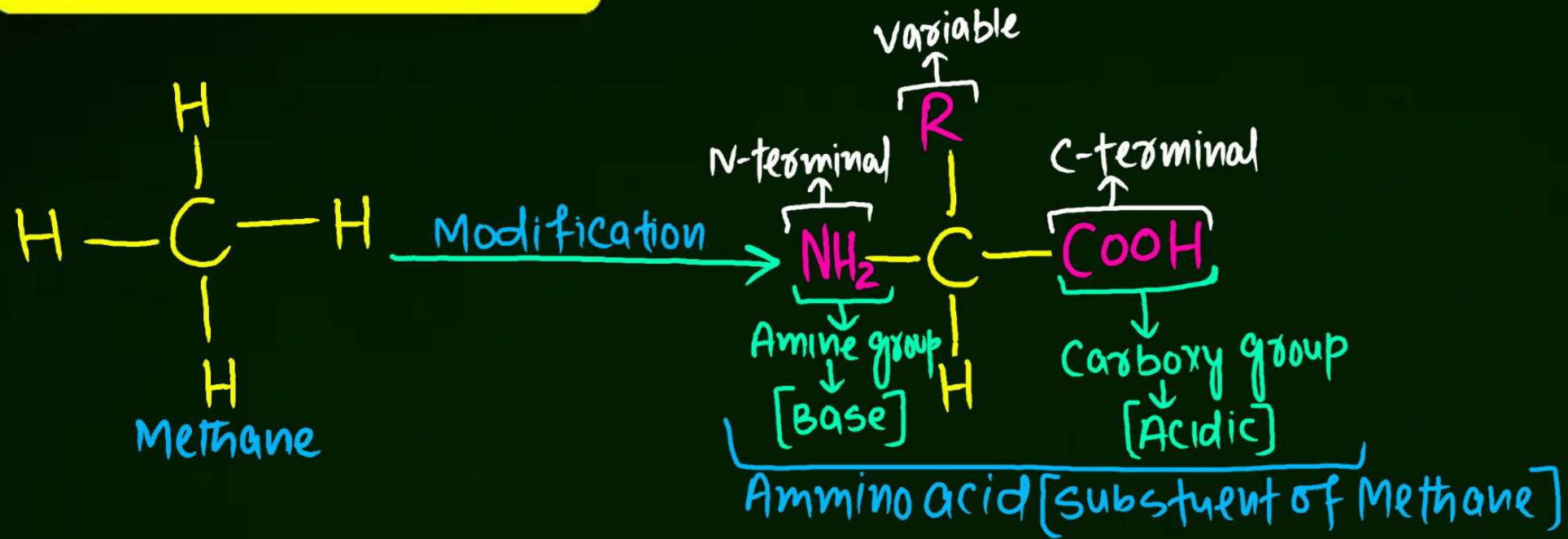


Physics Wallah

# Topics To Be Covered

- 1 Amino Acids
- 2 Carbohydrates
- 3 Previous Year NEET MCQ'S

# Amino acids



# Amino acids

the physical and chemical property depends on no. of COOH group and NH<sub>2</sub> group.

- ① Acidic Amino acid: → More COOH [Acidic group] and less basic group [NH<sub>2</sub>] e.g. Glutamic acid
- ② Basic Amino acid: → More NH<sub>2</sub> [Basic group] and less acid group [COOH] e.g. Lysine
- ③ Neutral Amino acid: → Equal number of COOH [Acidic] & Basic [NH<sub>2</sub>] group e.g. Valine

• Some amino acids are aromatic e.g. tyrosine, phenylalanine, tryptophan

Amino acids can be essential & Non-essential

Essential a.a

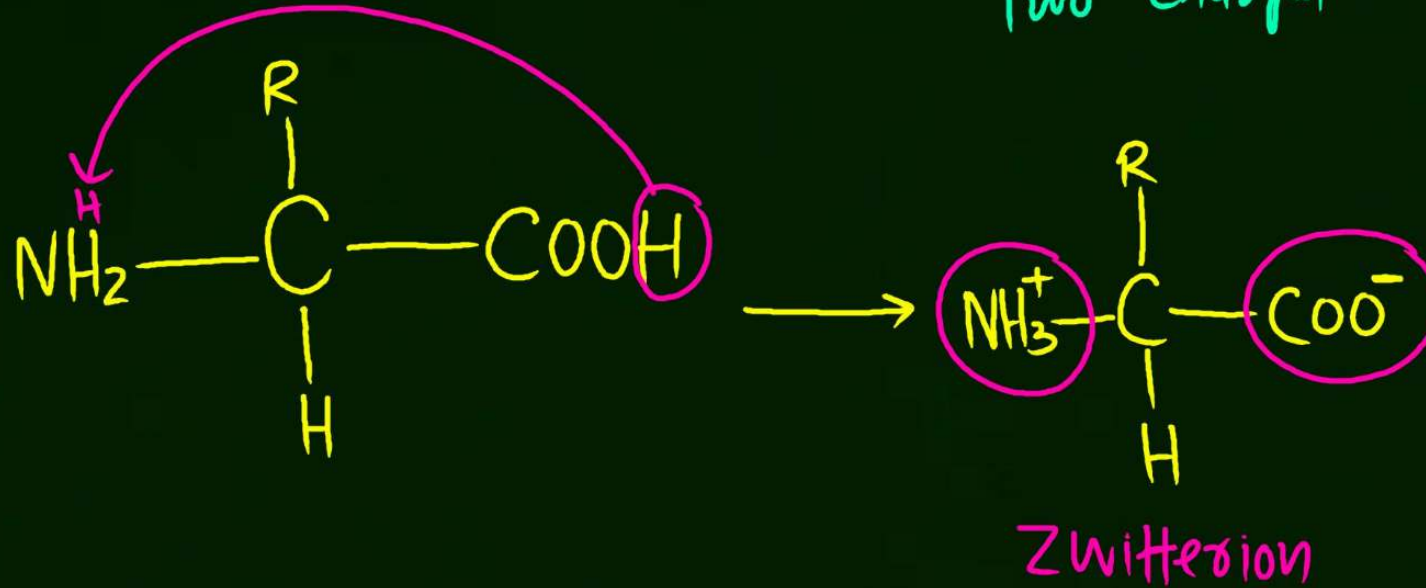
Can't be made by the body and must be taken from diet.

Non-essential a.a

Can be made by the body.

# Amino acids

- Amino acids acts as Zwitterion [German word → Zwitter ion  
Two charged]



- Zwitterion is a molecule that carries both positive and negative charges making it a neutral molecule.
- It is formed when
  - Carboxylic group loses proton  $[COO^-]$
  - the amine group gains proton  $[NH_3^+]$

# Carbohydrates / Saccharides

- It is an organic compound having C, H, O in a ratio of 1:2:1
- present in foods like Rice, fruits.



## Types of Carbohydrates

### Monosaccharides

Single + Sugar

- Which is made of only one monomeric unit.
- Hydrolysis is not possible.
- Example Glucose, Fructose.

### Oligosaccharides

Few Sugar

- Which is made of 2-9 monomeric units / monosaccharides.
- Two monosaccharides → Disaccharide
- Three monosaccharides → Trisaccharide
- 4 monosaccharides → Tetrasaccharide
- ⋮
- So on upto 9 → Nonasaccharide

### Polysaccharides

Many Sugar

- Which is made of 10-10000 monosaccharides
- Not sweet in taste and not known as sugars.
- Right side of polysaccharide is reducing end while left side is non-reducing.

# Polysaccharides



## Types of Polysaccharide

### Homopolysaccharide

Same

- Same type of monosaccharides (e.g.)

- ① Cellulose: → Present in plant cell wall
  - Most abundant organic compound on earth
- ② Chitin: → 2nd most abundant organic compound on earth
  - Present in Fungal cell wall & Skeleton of Arthropoda.
- ③ Glycogen: → As stored food in Animals.
- ④ Starch: → As stored food in plants.
- ⑤ Inulin: → used in detection of GFR [kidney function]
  - It is a polymer of Fructose

### Heteropolysaccharide

Different

- Different types of monomers are present (e.g.)

- ① Heparin
- ② Pectin
- ③ Agar
- ④ Hemicellulose
- ⑤ Hyaluronic acid
- ⑥ Peptidoglycan.

## Question [NEET-2020]

## Kaun Banayga Nabkaar



Identify the basic amino acid from the following.

1 Tyrosine

2 Glutamic acid

3 Lysine.

4 Valine

## Question [NEET-2023]

Kaun Banayga Nabkaar



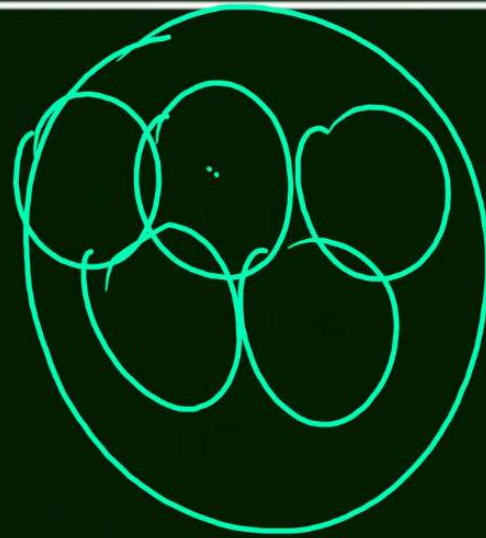
Inulin is a polymer of?

**1** Fructose.

**2** Glactose

**3** Amino Acid

**4** Glucose



## Question [NEET-2023]

## Kaun Banayga Nabkaar



Cellulose doesn't form blue colour with iodine because:

**1** It doesn't contain complex helices and hence can't hold iodine molecules. ✓

**2** It breaks down when iodine reacts with it ✗

**3** It is a disaccharide ✗

**4** It is a helix molecule ✗

**Thank**  
*You*